A Network and Its Reflection: What True, False, and Noise Self-Model Broadcasts Do and Do Not Change in an Evolving Multi-Agent Network

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Abstract. Many networks respond to measurements of their own state: markets to indices, organizations to metrics, platforms to trending signals. We present a controlled laboratory for this reflexive loop: an evolving agent population forms an adaptive network whose macrostate is compressed into a self-model and broadcast back — accurately, distorted, or as matched-range noise. Across 1,227 deterministic runs with paired-seed comparisons: measurement without broadcast is causally inert. A natural self-fulfilling-prophecy statistic registers strong apparent attraction toward false and replayed self-models — but a passive counterfactual, scoring untreated twins against matched references, fully accounts for it: a caution for empirical claims of performativity. What broadcasts demonstrably change is behavior and structure, and which lies come true is set jointly by the lie and the response regime. Finally, heritable attention to the broadcast declines across generations, most strongly under high-drive false and noisy feedback: populations evolve to ignore consequential misleading observers.

Keywords: adaptive networks, agent-based modeling, reflexivity, self-fulfilling dynamics, performative prediction, evolution of trust

1 Introduction

Adaptive networks — graphs whose topology coevolves with the dynamical states of their nodes — are a central object of network science [1, 2]. A distinctive subclass has received far less controlled attention: networks that respond to *descriptions of themselves*. Financial markets move on published indices computed from their own trades [3]; universities restructure around rankings derived from their own behavior [4]; recommendation platforms amplify items *because* an internal measurement declared them trending. In each case a macroscopic measurement is compressed, published, and re-enters the system as an input to microscopic decisions — a loop Merton called the self-fulfilling prophecy [5] and whose measure-distorting variant is known as Goodhart's law [6].

Empirical study of this loop is confounded: in real systems one can rarely withhold the measurement, falsify it in controlled directions, or replay history

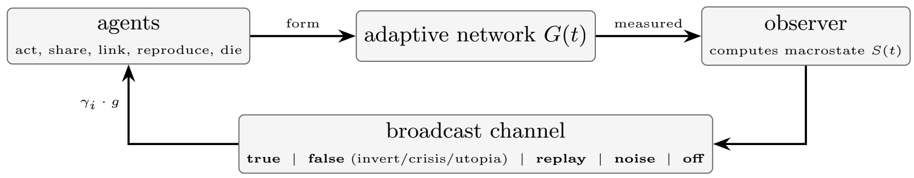

Fig. 1. The reflexive loop and its corruption points. Local actions generate the network; the observer compresses it into a self-model; the broadcast channel returns it truthfully, distorted, replayed from another run, as noise, or not at all. Action-propensity responses are scaled by heritable sensitivity γ_i and global gain g; the hub-targeted pruning probability is scaled by γ_i alone.

under identical randomness. We therefore built a minimal simulated world where the loop can be opened, closed, and corrupted at will, every run is exactly reproducible from (configuration, seed), and agents are transparent trait vectors rather than learned policies, so every pathway from broadcast to response is inspectable.

Our contributions are: (1) an open, deterministic experimental platform coupling an evolving agent ecology, an adaptive interaction network, a global observer, and a controllable broadcast channel; (2) a six-condition causal design — verifying observer-side-effect invariance and isolating the effect of broadcasting the measurement, with matched-range noise and replayed-self-model controls — together with a *passive-counterfactual methodology* for reflexivity metrics: a natural "story pull" statistic registers strong apparent steering that untreated twin trajectories fully account for — a direct caution for empirical tests of performative prediction; (3) evidence for what broadcasts *do* change: truthful feedback doubles costly cooperation, distorted broadcasts mobilize or pacify depending on the lie–response pairing (reversing under conformist responses), and broadcasts densify the network while false and noise feedback reduce hub concentration; and (4) we model receiver sensitivity as a continuous heritable trait and show that *its population distribution shifts systematically under feedback*, strongly tracking the broadcast-implied corrective-drive intensity and surviving a control that holds reproductive opportunity invariant — trust in the observer is not updated by any rule, it evolves. Figure 1 summarizes the design.

2 Related Work

Adaptive and temporal networks. Coevolution of state and topology is surveyed in [1, 2]; ours belongs to this family (births, deaths, rewiring by agent decisions). Robustness of networks to random and targeted node removal is classical [7]; we use targeted hub removal as a standardized perturbation. **Global coupling.** A mean field fed back to units reshapes collective dynamics [8]; here that field is multidimensional, potentially *false*, and units are born, die, and evolve. **Reflexivity and reactive measurement.** Self-fulfilling prophecy [5], reflexivity in markets [3], and the reactivity of rankings and metrics [4, 6] motivate our false-broadcast conditions; here they are dosable and ablatable. **Performative prediction and endogenous signals.** Predictions that alter the distribution they forecast are formalized as performative prediction [19, 23], including networked multi-agent settings [20] and populations that decide whether to trust predictions of collective outcomes [21] — there trust is *updated* from past accuracy, whereas here attention is inherited and mutated with no rule referencing reliability: trust is not updated, it evolves; self-fulfilling signals of an endogenous network state arise in congestion games [24]. Causal identification of performative effects without randomization is itself an active problem [22]; our contribution is narrower and complementary — we exhibit a concrete false-positive failure mode of directional-alignment metrics and diagnose it with paired passive twins in an evolving network (Sect. 5.2) — plus population turnover and a heritable receiver-sensitivity trait absent from these models. **Signaling and the evolution of trust.** In sender–receiver games receivers can evolve to ignore unreliable signals [9]; here the signal is *self-referential* and attention to it is a continuous heritable trait in an energy economy, not a payoff matrix. **Agent-based modeling.** Studies of collective opinion and cooperation are a mainstay here [16, 17]; our methodology follows the generative ABM tradition [10, 11] and the engine uses ideas from Mesa [12] and NetworkX [13]. Multi-agent reinforcement learning treats global state as an *architectural* aid [14]; we treat it as an *experimental variable* with controlled veracity. Populations of language-model agents are increasingly used to study emergent social behavior [15]; the present rule-based setting provides causally clean baselines for the shared-dashboard designs natural there.

3 The Model

3.1 Agents and ecology

Each agent i carries a heritable trait vector $\theta_i = (\mathrm{coop}, \mathrm{expl}, \mathrm{risk}, \mathrm{soc}, \mathrm{share}, \gamma_i) \in [0,1]^6$, where γ_i is its *global sensitivity* — how strongly broadcasts modulate its behavior. Agents hold energy; each step they pay metabolism and choose one action — harvest, share (costly transfer to a neighbor), connect, prune, reproduce, or idle — with probabilities proportional to trait- and state-dependent propensities. The resource pool regenerates logistically, giving density-dependent competition; reproduction requires an energy threshold and copies traits with Gaussian mutation ($\sigma = 0.05$); energy exhaustion or age causes death. Founding ages are staggered to avoid cohort artifacts.

3.2 Observer and broadcast conditions

Every $\Delta t = 10$ steps the observer computes the macrostate $S(t)$: population, fragmentation (1 minus largest-component share), degree- and betweenness-concentration, Freeman centralization [18], per-capita costly-helping rate, energy Gini, and turnover, plus mean traits (logged, never broadcast). The broadcast

Table 1. Model parameters (defaults; campaign overrides noted in the text).

initial population	150	observer interval Δt	10 steps
population cap	1200	feedback gain g	0.8 (swept 0.2–1.6)
initial edges per agent	3 (random)	mutation σ	0.05
initial energy	20	initial traits	mean 0.5, s.d. 0.15
resource capacity K	6000	reproduction threshold / cost	30 / 16
base regeneration	60/step	min. reproduction age	15 steps
regeneration rate r	0.12	lifespan (mean $\pm$ s.d.)	400 ± 80 steps
harvest cap	2.5/step	connect cost / share amount	0.25 / 3.0
metabolism	0.8/step	run length	3000–4000 steps
shock time t_{shock}	2000 (1500 in the 2× scale check)		

$b(t)$ carries five of these components, each normalized to $[0, 1]$: *fragmentation*, *degree concentration* (top-5% degree share; distinct from Freeman centralization), *cooperation*, *inequality*, and *turnover*. Six conditions define what agents receive (Table 2). Distorted broadcasts come in three modes: *invert* ($b_k = 1 - s_k$ componentwise), *crisis* (fixed alarming values: fragmentation 0.9, degree concentration 0.9, cooperation 0.1, inequality 0.9, turnover 0.9), and *utopia* (reassuring: 0.05, 0.2, 0.9, 0.1, 0.2).

3.3 Response model

Each step, agent i selects one action a with probability proportional to

$$w_i(a, t) \;=\; w_a\big(\theta_i, x_i(t)\big) \;+\; \gamma_i\, g\, f_a\big(b(t)\big), \tag{1}$$

where $w_a(\theta_i, x_i)$ is the trait- and local-state-dependent base propensity, g is the global feedback gain, and f_a is the response rule. In the primary *corrective* regime agents repair reported deficits: $f_{\text{connect}} = \max(0,\, b_{\text{frag}} - 0.3)$, $f_{\text{share}} = \max(0,\, 0.5 - b_{\text{coop}}) + 0.5\max(0,\, b_{\text{ineq}} - 0.5)$, $f_{\text{harvest}} = \max(0,\, b_{\text{turn}} - 0.5)$, and $f_{\text{prune}} = 0.3\max(0,\, b_{\text{cent}} - 0.6)$. In the *conformist* regime agents imitate the reported norm instead: $f_{\text{share}} = b_{\text{coop}}$, $f_{\text{prune}} = 0.5\, b_{\text{frag}}$, $f_{\text{connect}} = \max(0,\, 0.6 - b_{\text{frag}})$, and $f_{\text{harvest}} = b_{\text{ineq}}$. Two further mechanisms act on *target selection* rather than action choice, in both regimes: while reported degree concentration exceeds 0.6, a pruning agent cuts its highest-degree neighbor (ties by id) with probability γ_i (g scales action propensities only, not this probability), else a uniform neighbor; and new links attach to a neighbor-of-a-neighbor with probability 0.7 when available (else a uniform non-neighbor), an explicit triadic-closure tendency behind the clustering in Sect. 3.4. All rule constants are fixed across all experiments; Table 1 lists the model parameters.

3.4 Emergent network structure

The emergent networks are dense, clustered, and heterogeneous: at $t = 2{,}000$ the giant component has $L \approx 1.9$–2.5, clustering 1.8–2.4× a matched Erdős–Rényi

Table 2. Experimental conditions. All start from matched parameter distributions with paired random seeds.

Condition	Observer	Agents receive	Role
A — Local only	off	nothing	baseline
B — Observed, blind	on	nothing	measurement control
C — True feedback	on	accurate $S(t)$	main treatment
F — False feedback	on	distorted $S(t)$	reflexivity probe
N — Noise feedback	on	uniform random, matched ranges	stimulation control
R — Replayed self-model	on	another run's genuine $S(t)$	structure control

null, and degree coefficient of variation 0.6–0.75; against the stricter degree-preserving (configuration-model) null most of the clustering excess traces to the degree sequence (ratios 1.0–1.6), so we make no small-world claim. Feedback acts on a small, dense, degree-heterogeneous clustered graph, not a homogeneous mixing population.

3.5 Reproducibility

The simulation is deterministic given (configuration, seed): one PRNG drives all behavioral choices and a separate dedicated stream generates stochastic broadcast signals, so constructing the noise signal never advances the behavioral PRNG; agents act in sorted order and a state hash certifies replay (identical hashes verified across machines and operating systems). All code, configurations, and analysis scripts are public.

4 Experimental Design

The standardized shock removes, in a single step at $t_{\text{shock}} = 2000$ (except in the 2× scale check: $t_{\text{shock}} = 1500$ in a 3,000-step run), the highest-degree living agents (ties broken by agent id): a fixed 10 hubs in the mild variant, the top $\max(1, \lfloor 0.4\,|V_{\text{live}}| \rfloor)$ in the harsh variant. Four outcome *families* were fixed before the first campaign: (1) post-shock recovery time — the first $t > t_{\text{shock}}$ at which the living population regains 90% of its pre-shock baseline, defined as its mean over $[t_{\text{shock}} - 500,\, t_{\text{shock}})$; (2) mean post-shock fragmentation; (3) per-capita costly-helping rate, averaged after a 500-step burn-in; (4) self-model correspondence, operationalized as *story pull*: with $s_k(t)$ the value of macrostate component k at observer tick t, $b_k(t)$ the broadcast value, and $\varepsilon = 0.02$,

$$P \;=\; \Big\langle\, \operatorname{sign}\big(b_k(t) - s_k(t)\big)\,\big(s_k(t{+}1) - s_k(t)\big)\,\Big\rangle_{(t,k)\,:\,|b_k(t) - s_k(t)| > \varepsilon}, \tag{2}$$

averaged over all qualifying (tick, component) pairs across the five broadcast components; $P > 0$ means the macrostate moves toward the reference where the two differ. The threshold ε makes P well defined under false and noise broadcasts (and undefined for C, whose broadcast equals the current state).

Table 3. Main paired comparisons (Wilcoxon signed-rank on per-seed differences; matched-pairs rank-biserial r; bootstrap 95% CI of the median difference). FL = flagship (50 pairs), HS = harsh shock (30 pairs). "adj." compares $\Delta P = P_{\text{actual}} - P_{\text{passive}}$ between conditions (Sect. 5.2).

Outcome	Comparison	Δmedian [CI]	r	p
all state outcomes (FL, HS)	B vs A	all differences $\equiv 0$	—	—
story pull, raw (HS)	F vs N	$+0.0066$ $[0.0037, 0.0095]$	0.89	1.7×10^{-6}
story pull, adj. (HS)	F vs N	$+0.0022$ $[-0.0003, 0.0034]$	0.36	0.084
story pull, adj. (FL)	F vs N	$+0.0004$ $[-0.0016, 0.0023]$	0.08	0.63
cooperation (FL)	C vs A	$+0.200$ $[0.172, 0.239]$	1.00	8.9×10^{-15}
cooperation (HS)	C vs A	$+0.209$ $[0.170, 0.248]$	0.99	9.3×10^{-9}
fragmentation (FL)	F vs A	-0.017 $[-0.034, -0.006]$	-0.53	9.5×10^{-4}
$\Delta\bar{\gamma}$ (HS)	F vs A	-0.336 $[-0.456, -0.281]$	-1.00	1.9×10^{-9}
$\Delta\bar{\gamma}$ (HS)	N vs A	-0.323 $[-0.431, -0.241]$	-0.94	2.0×10^{-7}
recovery time (FL, HS)	C, F, N vs A	Δmedian $= 0$	$\lvert r\rvert < 0.5$	> 0.22

Outcome pre-specification. The four families above were frozen before the first campaign. Eq. 2 is a *post hoc* directional refinement of the fourth (self-model correspondence), adopted because the original distance-based form is degenerate under truthful feedback; both yield the same raw F-vs-N ordering. No pre-specified outcome was dropped, and all other reported measures — including the evolved trait change — are exploratory.

Passive counterfactual. Crucially, P can also be evaluated on trajectories that never received the reference: for each treated run we compute P on the paired-seed *untreated* trajectory (A/B) scored against a matched reference of the same type, defining the *counterfactual performativity gap* $\Delta P = P_{\text{actual}} - P_{\text{passive}}$ (Sect. 5.2). Campaigns: a *flagship* run (conditions A/B/C/F_{invert}/N $\times$ 50 paired seeds, mild shock), a *harsh-shock* replication (40% hub removal, $\times$30 seeds), a *sensitivity sweep* (feedback gain $g \in \{0.2, 0.8, 1.6\} \times \{\text{A, C, F}_{\text{invert}}, \text{F}_{\text{crisis}}, \text{F}_{\text{utopia}}, \text{N}\} \times 20$ seeds), a 2$\times$ *scale check* (doubled population and resources), a conformist *response-regime campaign* (15 seeds per cell), and a *replay control* (30 receiving runs on the harsh-shock seed list plus 15 auxiliary source runs): the core campaigns comprise $250 + 150 + 360 + 90 + 32 + 30 + 15 = 927$ runs of 3,000–4,000 steps; the reproduction-neutral control (Sect. 5.5) adds 300, for 1,227 in total. Because conditions share seed lists the design is paired; the primary analysis is a Wilcoxon signed-rank test on per-seed differences with matched-pairs rank-biserial r and a bootstrap 95% CI on the median difference. Unpaired Mann–Whitney U with Cliff's δ is a secondary robustness view; all inferential claims follow the paired analysis, and other logged measures are exploratory.

5 Results

Table 3 summarizes the main paired comparisons; the following subsections unpack them.

5.1 Measurement alone is causally inert

Under paired seeds, condition B is trajectory-identical to A: every per-seed difference is exactly zero, on every outcome, in both campaigns. The instrument acquires causal power only when it broadcasts.

5.2 Apparent story pull is explained by a passive counterfactual

Except where stated, results concern the corrective regime; Sect. 5.4 shows how they change under conformity.

Evaluated naively, the story-pull statistic appears to show strong self-fulfilling steering. Raw P (harsh shock; medians, bootstrap 95% CIs) is positive in all three non-truth signal conditions — F 0.0078 $[0.0066, 0.0100]$, N 0.0014 $[0.0001, 0.0023]$ — and largest for the replayed genuine self-model of a sibling world (condition R; 30 receivers on the harsh-shock seed list, one of 15 source trajectories each): 0.0241 $[0.0197, 0.0265]$. The raw hierarchy R > F > N is highly reliable (F vs N $r = 0.89$, $p \approx 1.7 \times 10^{-6}$; robust to one receiver per source, $n = 15$, $r \geq 0.98$, and to a source-clustered bootstrap).

A short observation shows why raw P can mislead. Under the inverted reference $b_t = 1 - s_t$, $\mathrm{sign}(b_t - s_t) = \mathrm{sign}(\frac{1}{2} - s_t)$; if an untreated component simply mean-reverts toward mid-range, $s_{t+1} - s_t = -\kappa(s_t - \frac{1}{2}) + \eta_t$ with $\kappa > 0$ and $\mathbb{E}[\eta_t \mid s_t] = 0$, then $\mathbb{E}[\mathrm{sign}(b_t - s_t)(s_{t+1} - s_t) \mid s_t] = \kappa \, |s_t - \frac{1}{2}| > 0$: ordinary mean reversion masquerades as attraction toward the mirror, with zero causal feedback; post-shock relaxation plays the same role for the replayed reference, which tracks where recovering trajectories go anyway. The remedy is the *counterfactual performativity gap* ΔP: score the paired-seed *untreated* trajectory (A, or the identical B) against a matched reference of the same type, post hoc — mirror, replayed source, or noise sequence. Untreated worlds "move toward" these references *more* than treated worlds move toward their actual broadcasts — passive medians 0.0119 (invert), 0.0300 (replay), 0.0073 (noise) — giving uniformly negative ΔP: -0.0036 (F), -0.0058 (R), -0.0055 (N), each $p \leq 10^{-5}$. Between-condition contrasts of ΔP are not significant in either campaign (F vs N: harsh $p = 0.084$; flagship $p = 0.63$; R vs N: $p = 0.95$). Raw P is thus dominated by the intrinsic dynamics anticipated above, which align best with realistic references and worst with unstructured noise (excluding the shock transition changes medians by $< 2\%$). Response activation confirms the reading: F keeps four of five corrective rules active on $\geq 90\%$ of ticks while R is mostly quiescent ($\langle |b_k - s_k| \rangle$: 0.64, 0.36, 0.10 for F, N, R), yet R shows the largest raw pull: its large raw pull is not explained by response activation. Under the primary corrective regime, Eq. 2 therefore provides *no evidence of aggregate content-specific attraction beyond intrinsic dynamics*; the small negative ΔP, similar across signal types, is consistent with broadcasts mildly dampening that drift. This is the paper's methodological result: a statistic natural in empirical performativity studies registers large, internally consistent, and entirely artifactual "steering" unless checked against passive twins.

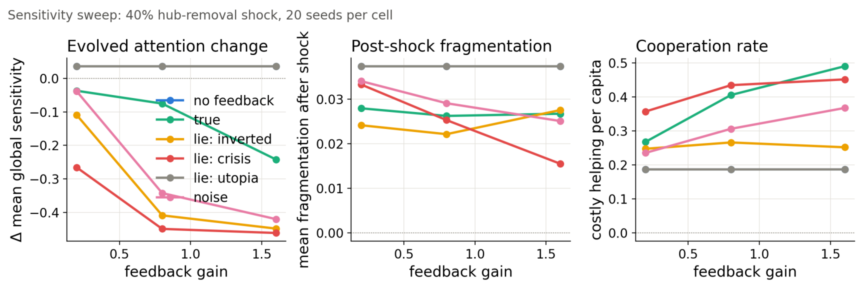

Fig. 2. Sensitivity sweep (40% hub-removal shock; 20 seeds per cell; medians). Left: evolved change in mean global sensitivity γ — attention to the self-model varies strongly with broadcast type and gain; the flattering (utopia) lie tracks the no-feedback baseline exactly. Middle: post-shock fragmentation — under corrective responses the alarmist (crisis) lie does *not* come true. Right: cooperation rate — deficit signals mobilize costly helping.

Structural spillovers, by contrast, are real. Direct outcome contrasts are unambiguous: agents persistently told the network is fragmenting over-connect it (mean degree 22.4 vs 11.2 in matched runs at $t = 2000$), reducing post-shock fragmentation (flagship paired Δmedian -0.017, $r = -0.53$, $p \approx 10^{-3}$; directionally consistent, ns, under harsh shock). Broadcasts densify and flatten the network: relative to A (paired, harsh shock, post-transient medians), mean degree rises $+1.1$ under truth ($r = 0.49$), $+3.8$ under the inverted lie ($r = 0.80$), $+2.8$ under noise ($r = 0.73$), while Freeman centralization (F: -0.026, $r = -0.74$; N: -0.010) and betweenness concentration (F: -0.007, $r = -0.67$; N: -0.008) fall: the fed-back network is denser and less hub-dominated than the silent one.

5.3 The type of lie sets what changes

The three distortion modes produce qualitatively different *outcome* regimes (Fig. 2). The *inverted* self-model, always reporting the mirror of reality, keeps most corrective responses persistently active (four of five on $\geq 90\%$ of ticks): it mobilizes costly helping ($+0.106$ over baseline, flagship $r = 0.82$) and drives the over-connection above. The *crisis* self-model is self-defeating for structure — corrections push fragmentation away from the alarmist report — yet its permanent cooperation-deficit signal mobilizes the highest costly-helping rates observed under correction (0.35–0.45 across gains). The *utopia* self-model is behaviorally inert: because responses fire only on reported deficits, a flattering broadcast triggers nothing and is indistinguishable from disconnecting the channel, at every gain: under purely corrective rules, flattery equals silence. Caveat: these first-order response *directions* follow from the rules (Eq. 1) by construction; not built in are the magnitude asymmetry across distortion types, the structural spillovers, and the evolutionary consequences below.

Table 4. The same lie under two response regimes (medians; harsh shock, $g = 0.8$). Baseline (A): cooperation 0.18, fragmentation 0.038.

Lie	Corrective		Conformist	
	coop.	fragm.	coop.	fragm.
Invert	0.27	0.021	0.42	0.054
Crisis	0.36	0.025	0.26	**0.400**
Utopia	0.19 (inert)	0.037 (inert)	**0.428**	0.025

5.4 The response regime decides which lies come true

Conformist agents reverse the taxonomy exactly as the mechanism predicts (Table 4). The *crisis* lie, self-defeating under correction, becomes strongly self-fulfilling under conformity: told the world is fragmenting, imitators cut ties and fragmentation rises an order of magnitude above baseline (0.400 vs 0.038). The *utopia* lie flips from behaviorally inert to benevolently self-fulfilling: "everyone cooperates" becomes true by imitation (cooperation 0.428). Under conformity even noise mobilizes: imitating randomly reported cooperation levels raises costly helping to 0.445, as high as any lie, since imitation amplifies whatever is reported. Which false self-models come true — in the outcome sense — is therefore a property of the lie–response pair, not of the lie alone. Compactly: with macrostate update $S_{t+1} = \Phi\big(S_t, R(B(S_t))\big)$ for broadcast map B and response rule R, self-fulfillment is a property of the composition $R \circ B$, not of B alone.

5.5 Attention declines under consequential broadcasts

Mean heritable global sensitivity $\bar{\gamma}$ changes over $\sim$40 generations as an emergent, unprogrammed function of broadcast quality (Fig. 3). At $g = 0.8$ (harsh shock): no broadcast -0.03 (neutral drift), truth -0.10, noise -0.34 ($r = -0.94$, $p \approx 2 \times 10^{-7}$), false -0.42 (Δmedian -0.336 $[-0.456, -0.281]$, $r = -1.00$, $p \approx 1.9 \times 10^{-9}$). The sweep is dose-ordered: the decline under the inverted broadcast deepens from -0.12 ($g = 0.2$) to -0.45 ($g = 1.6$) — and at the highest gain even *accurate* feedback shows a marked decline (-0.24). The categorical ordering compresses into one continuous relationship: with broadcast-implied corrective-drive intensity $I = g \, \langle \sum_a f_a(b(t)) \rangle$ (computable from the broadcast alone), evolved attention tracks I across the 15 sweep cells almost perfectly (Spearman $\rho = -0.99$; also strong at run level, $\rho = -0.83$, $n = 300$). Utopia ($I = 0$) is untouched; truth exhibits less decline where its corrective drive is smaller at equilibrium; unreliable high-drive broadcasts pay most. The association is consistent with corrective-drive intensity, rather than veracity alone, organizing the evolutionary response. Since I omits the per-agent factor γ_i (the quantity under selection) and is computed from realized rather than independently randomized trajectories, we treat it as descriptive, not as causal mediation.

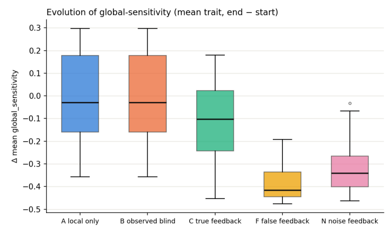

Fig. 3. Evolution of attention to the self-model (harsh shock, $g = 0.8$, 30 seeds per condition): change in mean $\bar{\gamma}$, first to final observer record. The ordering (false < noise < true < none) emerged from the evolutionary dynamics alone; nothing in the rules references broadcast reliability.

Not a reproductive-opportunity artifact. Feedback adds weight only to non-reproductive actions, so $P(\text{select reproduce} \mid \gamma_i, b) = w_r/(W_0 + \gamma_i g D(b))$ decreases in γ_i whenever $D(b) > 0$ — and I governs that penalty, so the correlation could be architectural. Rerunning the sweep (300 runs) with a reproduction-neutral variant rescaling the reproduce weight by $(1 + \gamma_i g D/N_0)$, holding that selection probability exactly at its no-broadcast value while leaving all non-reproductive feedback weights unchanged, leaves the decline essentially unchanged: median retention 98% across all twelve cells with a negative baseline $\Delta\bar{\gamma}$, no cell shifting significantly, $\rho = -0.98$ persisting, and truth declining *more* under the control. Selection on γ is thus ecological, not a reproductive-opportunity cost. The remaining pathway is not explained by a simple negative marginal fecundity gradient (within-run $\rho \approx +0.05$ between γ_i and offspring count under F) nor by shock mortality (-0.40 without the shock vs -0.39 with), and is left open.

5.6 Truthful feedback doubles cooperation; recovery time is untouched

True feedback roughly doubles costly helping (flagship: 0.397 vs 0.179; paired Δmedian $+0.200$ $[0.172, 0.239]$, $r = 1.00$, $p \approx 9 \times 10^{-15}$; replicated under harsh shock, $r = 0.99$), rising monotonically with gain ($0.27 \rightarrow 0.48$). The loop implements distributed corrective feedback: deficits reported at the population level are repaired by uncoordinated individual action. By contrast — a pre-specified null — we detected no evidence of a feedback effect on post-shock recovery time in any condition or campaign, even with 40% of hubs removed (all $p > 0.12$; median ≈ 40 steps): within the regimes tested, resilience is demographic, not informational.

5.7 Robustness to system scale

Re-running the core conditions (A, C, F_{invert}, N) at doubled population and resources (steady-state ≈ 120–130 vs ≈ 60; $n = 8$ pairs) preserves the main effects: cooperation under truth 0.399 vs 0.188 (Δmedian +0.151 $[0.119, 0.247]$, $r = 1.00$, $p = 0.008$); distrust ordering intact (F vs A -0.299 $[-0.388, -0.212]$, $r = -1.00$, $p = 0.008$; N vs A -0.219, $r = -0.89$, $p = 0.023$). The principal phenomena persist at $2\times$ scale.

6 Discussion and Limitations

As noted alongside the results, the first-order response directions are encoded in the rules; the findings live one level up. First, the methodological lesson: a statistic that looked like compelling evidence of self-fulfilling dynamics — positive, replicated, and largest for realistic signals — was fully explained by intrinsic dynamics on passive twin trajectories. Empirical tests of performativity face this identification problem [22], typically without access to the untreated twin our platform provides. Second, the substantive findings: broadcasts change behavior and structure with a regime-dependent taxonomy of lies, and above all the evolutionary layer, in which the credibility of a collective self-model becomes an evolving property of the population. Limitations: a single ecology and parameter family (though effects persist across an $8\times$ gain range and a $2\times$ scale change), fixed rule-based rather than learned response policies, and one network per run rather than interacting populations.

The results have a practical reading. Metric-driven infrastructure (autoscalers, observability pipelines, trending algorithms) is exactly this loop: the false-broadcast conditions model telemetry corruption, and under deficit-driven responses *flattering telemetry can be operationally equivalent to no telemetry* while raising no alarms. Episodes of reactive measurement such as institutional ranking reactivity [4] are natural empirical targets — with the passive-counterfactual check as the entry requirement. For multi-agent AI, where populations of language-model agents [15] make shared collective-state dashboards a natural design, the evolution-of-distrust result suggests persistent misreporting does not merely misdirect a population — it degrades the control channel itself.

7 Conclusion

We introduced a reproducible causal laboratory for reflexive self-modeling in adaptive networks and showed that (i) the measurement-only control is trajectory-identical to baseline, while broadcasting changes behavior, structure, and evolution; (ii) apparent self-fulfilling "story pull" — strong, replicated, largest for realistic signals — vanishes against a passive counterfactual, a caution for empirical performativity claims; (iii) which lies come true, in the outcome sense, is set by the lie–response pair, not the lie alone; and (iv) heritable attention to the collective self-model declines across generations, tracking the broadcast-implied corrective-drive intensity, making a population's trust in its own self-model an evolving quantity that high-drive misleading feedback can spend down. Future work: learned response policies and language-model agents.

Reproducibility. Code, configurations, analysis scripts, pinned dependencies, and deterministic regeneration of every run: `github.com/Ivan825/Project-ONE`, tag `cn2026-submission`; stored run records and replay trajectories ship as release assets there.

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